

BIGYAN BUDHA

Civil Engineering Student (Internship Applicant) | Hydropower · GIS · CAD

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PROFESSIONAL SUMMARY

Final-year Civil Engineering student at Khwopa Engineering College (Purbanchal University), graduating in 2027, with hands-on hydropower pre-feasibility experience in GIS catchment delineation, flow-duration analysis, hydraulic modeling and CAD layout. Seeking a Civil Engineering internship to apply analytical and drafting skills on hydropower, hydrology, GIS and construction projects.

EDUCATION

Bachelor of Civil Engineering Expected 05/2027

Khwopa Engineering College (Purbanchal University), Bhaktapur, Nepal

Relevant coursework: Hydrology & Water Resources Engineering, Hydropower Engineering, Engineering Surveying, Construction Project Management, Structural Engineering, Transportation Engineering, Geotechnical / Rock Engineering

+2 / Higher Secondary Education 01/2023

Khwopa Secondary School, Bhaktapur, Nepal

ACADEMIC PROJECT EXPERIENCE

Pre-Feasibility Study of Upper Tawa Khola Hydropower Project (6.5 MW) Ongoing

Final-Year Group Project (Team of 5) | Khwopa Engineering College

- Delineated the **91.06 sq. km** intake catchment from a DEM in ArcGIS Pro; analyzed rainfall data and calculated basin length, stream length, drainage density and time of concentration.
- Developed flow-duration curves from **4 reference hydrological stations** (602, 602.5, 684, 730), evaluated the PCAR and CAR methods, and adopted a design discharge of **3.77 m³/s**.
- Modeled hydraulic conditions in HEC-RAS using DEM data to calculate a final net head of **140.10 m** for the hydropower scheme.
- Prepared an AutoCAD layout plan consolidating the proposed **4.48 MW** hydropower scheme into a single drawing.

TECHNICAL SKILLS

Hydrology & Hydropower: catchment delineation, rainfall data analysis, flow-duration curves, design discharge estimation, net head calculation, headworks concepts

GIS, CAD & Modeling: ArcGIS Pro (DEM-based catchment delineation, rainfall analysis), AutoCAD (project and assignment plan layouts), HEC-RAS (hydraulic modeling)

Engineering Tools: MS Excel, quantity estimation, C programming

Learning / basic level: Civil 3D, Revit, SketchUp, Phase

TRAINING

AutoCAD Professional Training Course (30 hours), Khwopa Engineering College 09/2024 – 10/2024

ADDITIONAL INFORMATION

Award: First Place, Civil Engineering Department Quiz Contest, Khwopa Engineering College (09/2023)

Volunteering: Supported event management at the two-day National Exhibition “Aarohan 2.0,” Khwopa Architecture Students’ Activity Circle (07/2026)

Languages: English (Professional), Hindi (Working), Nepali (Native)